Does Cost-Efficiency Travel? What Agent Leaderboards Can Predict About Cost Across Workloads

Anonymous Author(s)
Affiliation
Address
email

Abstract

Agent leaderboards increasingly report task success and dollar cost, inviting users to choose a "cost-efficient" system from a single benchmark. We ask whether that recommendation travels to a new workload. Using public Holistic Agent Leaderboard (HAL) results, we hold the displayed HAL Generalist Agent scaffold fixed and compare repeated displayed model labels across six workloads. Raw dollar-cost ranks are strongly associated in several high-overlap pairs, but this signal is largely absorbed by stable displayed-label and benchmark cost propensity. Held-out prediction is strong for four workloads and uninformative for two lower-overlap workloads. Aggregate dollars per expected success are less stable: only three CostPerSuccess associations survive permutation testing and Holm correction, and the clearest raw-cost versus CostPerSuccess gap occurs for GAIA–SWE-mini. A separate matched public trace study gives the system-level context that leaderboard aggregates lack: across 977 exact task–model pairs, OpenHands uses about 15% fewer median turns and tool calls than SWE-agent, without a detected difference in resolved status, and the popular "expensive failure" premium dissolves entirely under exact task matching, indicating it measures workload composition rather than failing-run behavior. Taken together, the results show that raw cost is a useful but incomplete portability signal. It combines persistent label-level cost propensity with scaffold execution structure, and it should not be treated as a portable ranking of deployment efficiency.

1 The practical problem

Agent evaluation is expensive enough that cost has become a first-class leaderboard number. HAL reports 21,730 rollouts across nine benchmarks and roughly $40,000 in evaluation cost [Kapoor et al., 2026]. That makes a cost–accuracy frontier appealing: if two systems perform similarly, why not choose the cheaper one?

The hidden step in that reasoning is portability. A user sees that a label is inexpensive on one benchmark and assumes it will remain a good buy in a different environment. That assumption is plausible, but it is not innocuous. A reported dollar cost can reflect the model or provider, reasoning mode, prompt and tool setup, cache behavior, scaffold, and the workload itself.

The GAIA–SWE-mini comparison makes the issue concrete. Across 17 shared displayed labels, raw dollar-cost ranks are closely aligned (Spearman $\rho = 0.89$). But the ranking by aggregate dollars per expected success is much less aligned ($\rho = 0.29$); the paired raw-minus-CPS contrast is 0.60 with a 95% configuration-resampling sensitivity interval of [0.14, 1.13]. In practical terms, a GAIA-like leaderboard can offer a useful prior about raw spend for a SWE-like workload without providing a reliable ordering of expected dollars per successful task. This is an aggregate ranking example, not a claim about a named model or a production deployment.

We therefore ask three questions. First, when does an observed cost ranking predict a held-out workload? Second, does that ranking still travel after separating stable displayed-label cost propensity from benchmark cost level? Third, what can public traces say about the scaffold-level execution structure that aggregate leaderboards hide?

2 Why this is different from prior cost and benchmark work

This paper is an empirical evaluation study, not a new rank estimator. Kapoor et al. [2024] established the importance of reporting accuracy and cost jointly and motivates the frontier perspective used here. Ndzomga [2026] asks whether a smaller task subset preserves agent rankings within a benchmark; we instead ask whether an observed cost signal predicts a different workload. Benchmark^2 studies cross-benchmark rank consistency for static LLM benchmarks; our setting adds interactive scaffolds, dollar cost, success adjustment, and a held-out workload test. SWE-Effi examines resource effectiveness within software engineering; our result concerns the portability of public leaderboard signals across workloads. The matched Open-SWE extension does not claim to reproduce HAL cost measurements. It supplies separate, exact task–model evidence that scaffold choice can change observable trajectory burden.

Our primary data are the public `all_leaderboards_costs_HAL.csv` snapshot distributed with Ndzomga [2026]: 242 displayed evaluations across nine benchmarks, 11 scaffolds, and 29 displayed model labels. The coverage determines the design. Most scaffolds appear on one workload only, so the data do not support a general model–scaffold–workload decomposition. We instead fix the displayed HAL Generalist Agent scaffold, match exact public model-label strings, and retain six workloads with enough repeated labels: CORE-Bench Hard, GAIA, SciCode, ScienceAgentBench, SWE-bench Verified Mini, and TAU-bench Airline. This cohort contains 86 rows. USACO has one Generalist observation and is excluded.

The primary CSV supplies accuracy and aggregate dollar cost, but not token components, provider aliases, API calls, tool calls, retries, cache usage, or per-task trajectories. Accordingly, we do not call any adjustment "price controlled," and we do not infer token or failure costs from dollars. Most evaluations are single-run, so rollout-level uncertainty is unavailable.

For each pair with at least five shared labels, we rank the same shared displayed labels by each metric. A configuration-resampling sensitivity interval is obtained by drawing N shared labels with replacement, keeping each draw's paired values together, and recomputing the statistic; 10,000 draws yield its 2.5th and 97.5th percentiles. It describes sensitivity to the finite observed label set, not a rollout-level confidence interval. Raw-cost Holm correction applies to 15 tests. CPS has its own 20,000-draw label-pairing permutation tests and Holm correction over 15 tests.

$$\mathrm{CPS}_{i,b}(\epsilon) = \frac{C_{i,b}}{\max(A_{i,b}, \epsilon)},$$

where A is accuracy as a fraction and the primary floor is $\epsilon = 0.01$. The 5% and zero-exclusion versions are reported in the supplement. CPS is an aggregate dollars-per-expected-success proxy, not the mean cost of successful runs.

To test out-of-sample portability, we use leave-one-benchmark-out (LOBO) prediction. For each held-out workload, we estimate each label's centered log-cost propensity from the other workloads and predict the held-out cost rank. We also fit the descriptive pooled model

$$\log C_{i,b} = \mu + \alpha_i + \beta_b + \epsilon_{i,b}.$$

This 86-row model has 30 parameters and 56 residual degrees of freedom. Its fit is descriptive; LOBO is the stronger test of portability.

Finally, we report three decision frontiers: discrete nondominance, a HAL-inspired origin-anchored convex hull, and a 5% near-tie tolerance. These answer related but different decision questions.

3 Evidence

3.1 Raw cost predicts some held-out workloads

The most direct evidence comes from LOBO. A label's cost propensity learned from the other workloads predicts held-out raw cost ranks for CORE ($\rho = 0.78, N = 14, p = 0.0016$), GAIA ($0.87$, $N = 16, p < 0.0001$), SWE-mini (0.90, $N = 16, p < 0.0001$), and TAU (0.83, $N = 14, p = 0.0004$). These are not one-label artifacts: leave-one-label-out ranges remain positive for every omission, from [0.73, 0.91] for CORE to [0.88, 0.92] for SWE-mini.

The picture is not uniform. SciCode is unstable under leave-one-label-out analysis ([-0.50, 0.14]). ScienceAgentBench remains directionally negative ([-0.77, -0.43]), but it has seven labels and its original permutation result is not significant. SciCode has very low mean accuracy (2.1%) and one-third zero-accuracy labels; ScienceAgentBench has the narrowest log-cost dispersion. These are useful diagnostic facts, not a basis for a domain-level causal story.

3.2 What the raw-cost signal appears to contain

Among high-overlap pairs, raw dollar-cost correlations range from 0.74 to 0.93. A pooled benchmark-plus-label log-cost model has $R^2 = 82.0\%$ and adjusted $R^2 = 72.7\%$, but this in-sample fit should not be overread given 30 parameters and 86 rows. Its purpose is diagnostic. Once observed benchmark and displayed-label cost propensity are removed, none of the high-overlap residual rank associations survives a 20,000-draw permutation test.

The reasonable interpretation is not that cost is meaningless. Rather, much of the portable raw-cost signal is a stable observed label-level component. The public data cannot identify whether that component comes from provider pricing, reasoning settings, routing, token behavior, cache treatment, or other unrecorded setup choices.

3.3 Cheapness is not the same ranking as dollars per expected success

CPS changes the question from "which label is cheap?" to "which label buys expected success cheaply?" Under the 1% floor, CPS association survives 20,000-draw permutation testing and Holm correction for CORE–GAIA ($\rho = 0.87$, 95% sensitivity interval [0.63, 0.96]), CORE–TAU (0.83, [0.31, 1.00]), and GAIA–TAU (0.78, [0.42, 0.94]). Most other CPS associations remain uncertain.

The high-overlap mean CPS correlation is 0.59, compared with 0.84 for raw cost, but that average is descriptive rather than a population-level contrast. The clearest paired difference is GAIA–SWE-mini: raw cost exceeds CPS correlation by 0.60, with a 95% resampling interval [0.14, 1.13]. In short, a raw-spend ranking can travel while a success-adjusted ranking does not.

3.4 There is no universal cost–accuracy winner

No broadly tested label lies on every frontier under nondominance, the HAL-inspired hull, or the 5% tolerance rule. This is a practical guardrail, not the paper's central novelty. The stronger statement that no label appears on more than 4/6 frontiers fails under tolerance: o4-mini Low appears on 5/6.

4 A separate matched trace study

The leaderboard analysis tells us what travels, but not what happens inside an agent run. HAL public trace archives are encrypted. We therefore use a separate public Open-SWE-Traces boundary sample: four documented MiniMax-M2.5 shards under SWE-agent and OpenHands. It is not a random sample of the full corpus. We exactly match task ID and model, exclude duplicate task/model/scaffold records, and obtain 977 pairs.

The traces expose resolved status, structured messages, tool calls, and text. They do not expose billing, tokens, provider usage, retries, or latency. We therefore call turns, tool calls, and message characters trace-burden proxies, not cost.

At fixed task and model, OpenHands/SWE-agent has a median ratio of 0.85 for trajectory turns (10,000-bootstrap interval [0.82, 0.87]) and 0.85 for tool calls ([0.82, 0.86]). OpenHands has lower values on 69.7% of pairs for each measure. Total trajectory characters have a smaller ratio effect, 0.95 ([0.92, 0.98]), with OpenHands lower on 55.1% of pairs. Resolved status has no detected paired difference ($p = 0.197$). Thus, scaffolds can change visible execution burden even when task and model are held fixed. This does not show lower token use or dollar cost, and it does not identify the mechanism behind HAL rankings.

The same matched design tests the popular "expensive failure" heuristic. Across tasks, failed trajectories look 19–36% heavier than resolved ones, matching prior resource-effectiveness claims. But among the 107 pairs where the two scaffolds disagree on the identical task, the failed side is heavier only 47–52 times out of 107 on every metric ($p \geq 0.25$), and after normalizing by each scaffold's median burden, failed/resolved median ratios are 0.98–1.06 with intervals spanning 1. Hard tasks make agents both burn more and fail; unconditioned failure-cost ratios measure workload composition, not failing-run behavior, and inherit exactly the composition problem documented above for raw aggregate dollars.

5 Interpretation and implications

The evidence supports a narrower conclusion than a single leaderboard recommendation suggests. Raw dollar cost can be predictive across some workloads. In the HAL cohort, that prediction is robust for four held-out workloads and is not driven by a single displayed label. But the signal is partly stable label-level propensity, not a demonstrated measure of portable execution efficiency. Success adjustment changes the ranking and is only partly portable. The matched trace study supplies a compatible systems-level explanation: scaffold choices alter observable work even when task and model are fixed.

For practitioners, this means that a leaderboard's dollar figure is useful as a first screen, not as a final procurement decision. A target evaluation should include a success-adjusted metric and, where possible, trajectory telemetry. For benchmark builders, the missing fields are concrete: model/provider aliases, input/output/cache usage, calls, tool activity, retries, outcome labels, and per-task costs would make it possible to distinguish price from execution behavior.

6 Limits

This is a secondary-data analysis. Public display labels do not fully identify prompts, tools, budgets, harness versions, or routing. The HAL cohort is sparse and unbalanced, with mostly single-run rows. The public HAL snapshot cannot identify failure-cost ratios, retry behavior, token trajectories, tail cost, or a causal model–scaffold–workload decomposition. CPS is not task-level successful-run cost. The Open-SWE extension is external to HAL and uses a reproducible four-shard boundary sample, not the full corpus. Its proxies are not tokens or dollars. These limits define the claims rather than invalidate them.

7 Conclusion

A cost-aware leaderboard is better than an accuracy-only leaderboard, but it is not a portable deployment-efficiency ranking by default. In these public agent data, raw cost can generalize to some held-out workloads, yet it partly reflects stable displayed-label propensity and does not always preserve the ranking by dollars per expected success. Matched traces show that scaffold design changes observable execution burden at fixed task and model. The next generation of agent leaderboards should make these distinctions measurable rather than leaving them hidden inside one dollar number.

References

Sayash Kapoor et al. AI agents that matter. *arXiv preprint arXiv:2407.01502*, 2024. URL `https://arxiv.org/abs/2407.01502`.

Sayash Kapoor et al. Holistic agent leaderboard: The missing infrastructure for AI agent evaluation. *arXiv preprint arXiv:2510.11977*, 2026. URL `https://arxiv.org/abs/2510.11977`.

Franck Ndzomga. Efficient benchmarking of AI agents. *arXiv preprint arXiv:2603.23749*, 2026. URL `https://arxiv.org/abs/2603.23749`.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state that the analysis is descriptive, restricted to repeated public HAL display labels under a repeated displayed scaffold, and does not identify causal model or scaffold effects.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 6 discusses sparse coverage, display-label ambiguity, single-run evaluations, price dependence, missing token totals, limited domain comparisons, and frontier sensitivity.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper presents an empirical secondary analysis and no theoretical results.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Sections 3–4 and the appendix specify the public source, frozen hash, inclusion criteria, matching unit, frontier definitions, overlap threshold, resampling procedure, and seed.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: An anonymized supplement will contain the analysis code, provenance manifest, and reproduction instructions; the underlying public data source is cited.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: No models are trained. The data filtering, rank metrics, frontier definitions, resampling count, and seed are specified in Sections 3–4.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: We report configuration-resampling sensitivity intervals and explicitly state that they are not rollout-level or population-model confidence intervals and are not used for hypothesis tests.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [N/A]

   Justification: The work is a lightweight secondary analysis of public tabular data and uses no model training or agent rollouts.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The work uses public aggregate evaluation data, attributes source authors, reports limitations, and makes no claims about people or protected groups.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: We use **Yes** here because the work has practical consequences. Better reporting of success-adjusted cost and execution telemetry can reduce wasted evaluation and deployment spend. Conversely, treating a single leaderboard's dollar ranking as a deployment recommendation can misallocate resources or favor systems whose price/configuration signal does not transfer. The paper's safeguards are explicit claim boundaries, public-data provenance, and reproducible sensitivity analyses.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: The paper releases no trained model, agent capability, or new high-risk dataset.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: The data and methodological sources are cited; the supplement records the upstream repository, commit, source path, and declared MIT license.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The analysis code and generated tables are documented by a source manifest, artifact manifest, tests, and reproduction commands.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The study involves no crowdsourcing and no human subjects research.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The study involves no human subjects research.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

    Answer: [N/A]

    Justification: The core method is deterministic analysis of public tabular data; no LLM is used as a component of the empirical method.